Shashank Kumar

Full-Stack & Applied AI Engineer | AI Safety | LLM Systems
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Summary

Engineer with five years shipping production software, now working across applied cryptography and LLM systems. Author of CHRONOS, a Rust implementation of an ephemeral agent combining homomorphic encryption, a verifiable delay function, and a Groth16 erasure proof (8,267 R1CS constraints, 128-byte proofs verifying in 1 ms). Delivered a pre-launch security audit that surfaced a PCI-DSS violation and an authentication bypass. Rated top 5% on Alignerr's adversarial evaluation programme; Top Rated on Upwork. Owns problems end to end: design, Rust and Python implementation, benchmarking, and precise reporting of what is and is not proven.

Technical Skills

Languages: Rust, Python, TypeScript, JavaScript, C++, C#, SQL, Bash

Cryptography: Groth16 zk-SNARKs (arkworks), Poseidon hash, verifiable delay functions, BLS12-381, ML-DSA, AES-256-GCM, authenticated encryption

AI / ML: Multi-agent orchestration (FSM/DAG), RAG pipelines, prompt engineering, LLM fine-tuning (LoRA, DPO), RLHF, adversarial red-teaming, formal verification (Z3 SMT), graph-based agent analysis (FalkorDB)

Backend: FastAPI, Node.js, Express, GraphQL, gRPC, PostgreSQL, pgvector, TimescaleDB, Redis, MongoDB, REST APIs

Frontend: Next.js, React, Tailwind CSS, HTML5, CSS3

Infrastructure: Docker, GitLab CI/CD, GitHub Actions, Ansible, AWS, Azure, Prometheus, Grafana, Hugging Face, OpenAI and Anthropic APIs

Security: PCI-DSS auditing, threat modelling, mTLS, JWT, HMAC-SHA256, secrets management

Experience

Community Member - Cohere (Remote) Sep 2026 – Present

- Admitted to the Cohere Open Science Community based on a technical review of open-source verifiable AI projects.

Full Stack AI Engineer - BitNorm (Netherlands, Remote) Nov 2025 – Present

- Security-audited a pre-launch Rust and GraphQL platform; identified critical vulnerabilities in payment handling, authentication, and secrets management, including a PCI-DSS compliance violation.
- Delivered a prioritised remediation plan accepted by the engineering team: payment tokenization, GraphQL field masking, secrets management, and CORS hardening.
- Diagnosed database query inefficiencies in nested GraphQL resolvers and specified a batching pattern to reduce per-request round-trips.

AI Systems Evaluator and Red Teamer - Alignerr (Remote) Apr 2025 – Present

- Rated **top 5%** (platform-issued badge) across 1,000+ adversarial evaluation tasks.
- Authored 500+ adversarial red-team prompts against pre-release frontier models, targeting jailbreak resistance, factual grounding, and instruction-following.
- Evaluated three or more pre-release LLMs for factual accuracy and logical coherence ahead of production release.

Freelance AI and Full-Stack Developer - Upwork (Remote) May 2024 – Present

- Delivered LLM safety evaluation and adversarial testing for a client platform; **client-measured 25% improvement** in their internal safety scoring.
- Built a healthcare support chatbot on GPT-4 and Flask covering 500+ patient FAQs; 92% response accuracy and an 80% reduction in response time.
- Built custom RAG pipelines improving answer accuracy on client document sets; maintained **Top Rated** status on the platform.
- Annotation and model-evaluation work via Outlier AI (2024–present) alongside development engagements.

Selected Projects

CHRONOS - Ephemeral AI Agent with Verifiable Self-Termination Rust, arkworks, TFHE-rs

- Composed homomorphic encryption, a Wesolowski verifiable delay function, and a Groth16 erasure proof so an agent's key is released only by sequential work and its destruction is checkable by a third party. Hand-built Poseidon-128 over BN254 and a Poseidon-based AEAD for in-circuit authenticated decryption.
- Measured: **8,267 R1CS constraints** (all load-bearing), **128-byte proofs verifying in 1 ms**, VDF throughput constant in T at $\sim$**505k squarings/sec**; containment invariants checked exhaustively over 1,728 abstract states at startup.
- Documented limitations: single-party trusted setup, and FHE inference at small scale only (2.29 s per homomorphic multiplication). github.com/ChronosResearch/project-chronos

PAAC - Provably Aligned AI Core Python, Z3 SMT

- Deterministic safety wrapper using the Z3 SMT solver to verify constraints on self-modifying agent code before execution rather than after. **382 passing tests** across the constraint layer, solver integration, and failsafe paths. github.com/ChronosResearch/paac

Swarmchaser - Coordinated AI Agent Detection Python, MinHash, Sep 2026

- Detects coordinated swarms of AI agents from public post data using key extraction, MinHash paraphrase clustering, and statistical steering tests. Identified 119 swarms across 125,885 posts from the Moltbook Observatory Archive. **Open source.** github.com/sidthebuilder/swarmchaser

Provenant - Agent Swarm Detection and Block-Replay Engine Python, FalkorDB; Graph Hacks, Oct 2026

- Graph-based detection and "what-if" replay for coordinated AI agent swarms. Detected 117 swarms across 125,885 posts in 7 s; block replay forks the graph to measure collateral before applying a ban. Shared-memory benchmark on 180k+ real AI Village messages achieved 52.3% recall@20. **16 tests pass.** github.com/sidthebuilder/provenant

PyRobotics - Robotics Algorithm Reference Library Python, NumPy

- Ten-plus module library implementing RRT*, A*, LQR, MPC, MPPI, EKF, UKF, particle filter, pose-graph SLAM, and a policy-gradient agent with a pure-NumPy neural network. Simulation only; tests cover simulation, perception, and control. github.com/sidthebuilder/PyRobotics

Preprints

None of the following are peer-reviewed. Full list: Google Scholar.

- *Project CHRONOS: A Fully Homomorphic Ephemeral AI Agent* - SSRN 6950898, Jun 2026.
- *The Provably Aligned AI Core: A Deterministic Safety Wrapper* - SSRN 6879218, Jun 2026.
- *The Zero-Training Enterprise Brain: A Live Semantic Reasoning Architecture* - SSRN 6842778, May 2026.
- *Building Epistemic Infrastructure for Enterprise AI Search* - SSRN 6585978, Apr 2026.
- *Beyond Single Prompts: Engineering a Multi-Agent Workgroup for End-to-End Tabular Data Science* - technical report, Mar 2026.

Education

B.Sc. Computer Science - University of the People Apr 2025 – Expected 2027
In progress. Coursework in algorithms, data structures, and software engineering.

B.A. (English, History, Hindi) - Swami Vivekanand Subharti University, Meerut 2021 – 2025

Additional

Certifications: CS50 Introduction to Computer Science (Harvard, edX, 2024) • Google Analytics and Digital Marketing (Google Digital Garage, 2023) • Scale AI Annotation Specialist

Languages: English (fluent), Hindi (native)

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